Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VI

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AIM: Tree Traversal: Write a program to:

a. Implement pre-order,
b. in-order,
c. Post-order traversal of a binary tree.

CODE:

data structure 6.py - C:/Users/mvluc/Documents/data structure 6.py (3.14.6)

File Edit Format Run Options Window Help

```python
class Node:
    def __init__(self,item):
        self.left=None
        self.right=None
        self.val=item
def inorder(root):
    if root:
        inorder(root.left)
        print(str(root.val)+"->",end='')
def postorder(root):
    if root:
        postorder(root.left)
        postorder(root.right)
        print(str(root.val) + "->", end="")
def preorder(root):
    if root:
        preorder(root.left)
        preorder(root.right)
        print(str(root.val) + "->", end="")
root = Node(1)
root.left = Node(2)
root.right = Node(3)
root.left.left = Node(4)
root.left.right = Node(5)
print("Inorder traversal")
inorder(root)
print("\npostorder traversal")
postorder(root)
print("\npreorder traversal")
preorder(root)
```

OUTPUT:

IDLE Shell 3.14.6

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```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit
MD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
============= RESTART: C:/Users/mvluc/Documents/data structure 6.py ============
Inorder traversal
4->2->1->
postorder traversal
4->5->2->3->1->
preorder traversal
4->5->2->3->1->
>>
```

Curiosity Questions:

Instructions:
For each of the following binary trees, write the:

1. **Pre-order Traversal** (Root → Left → Right)
2. **In-order Traversal** (Left → Root → Right)
3. **Post-order Traversal** (Left → Right → Root)

Tree 1

```
      A
     / \
    B   C
   / \   \
  D   E   F
```

Answer:

- Pre-order:
- In-order:
- Post-order:

Tree 2

```
       10
      /  \
     5    15
    / \   / \
   2   7 12 20
```

Answer:

- Pre-order:
- In-order:
- Post-order:

Tree 3

```
        M
       / \
      G   T
     /   / \
    D   P   Z
       /
      N
```

Answer:

- Pre-order:
- In-order:
- Post-order:

Tree 4

```
       50
      /
    30
   /  \
 20   40
        \
        45
```

Answer:

- Pre-order:
- In-order:
- Post-order:

Tree 5

```
        A
       / \
     B    C
    /    / \
  D    E    F
      / \
     G   H
```

Answer:

- Pre-order:
- In-order:
- Post-order: